Message Intelligence System

Part A: Probability & Conceptual Foundation (Theory)

1. What is Conditional Probability?

Conditional probability is the probability of an event A happening given that another event B has already happened: P(A|B) = P(A and B) / P(B). Example: P(Spam | message contains a URL) is the chance a message is spam when we already know it contains a URL.

2. Explain Bayes' Theorem and its importance in classification problems.

Bayes' Theorem is P(A|B) = P(B|A) x P(A) / P(B). It updates a prior belief using new evidence to give a posterior probability. In classification it becomes P(Class|Features) = P(Features|Class) x P(Class) / P(Features), and the model predicts the class with the highest posterior. It is the foundation of the Naive Bayes classifier.

3. What assumptions does the Naive Bayes Classifier make?

1) Features are conditionally independent of each other given the class. 2) Every feature contributes equally and independently to the outcome. 3) For Gaussian Naive Bayes, numeric features follow a normal distribution within each class. It is called naive because real features, such as message length and special characters, are often correlated.

4. Explain the working principle of KNN and SVM.

KNN: A distance-based method that stores the training data. For a new message it finds the K closest training points using a distance metric (such as Euclidean) and assigns the majority class among them. Features must be scaled, and K controls how smooth the decision is.

SVM: A margin-based method that finds the hyperplane separating the two classes with the maximum margin. The points closest to the boundary are the support vectors. Kernels (RBF, polynomial) allow non-linear boundaries, and C balances a wide margin against misclassification.

5. Compare distance-based, probabilistic, and margin-based classifiers.

Distance-based (KNN): classifies by closeness to neighbours. Simple, no training step, but sensitive to scaling and noise, and slow on large data.

Probabilistic (Naive Bayes): classifies by the highest class probability using Bayes' Theorem. Fast, works with little data and gives probabilities, but relies on the independence assumption.

Margin-based (SVM): classifies using the maximum-margin boundary. Strong when classes are well separated and works well in many dimensions, but is slower to train and harder to interpret.